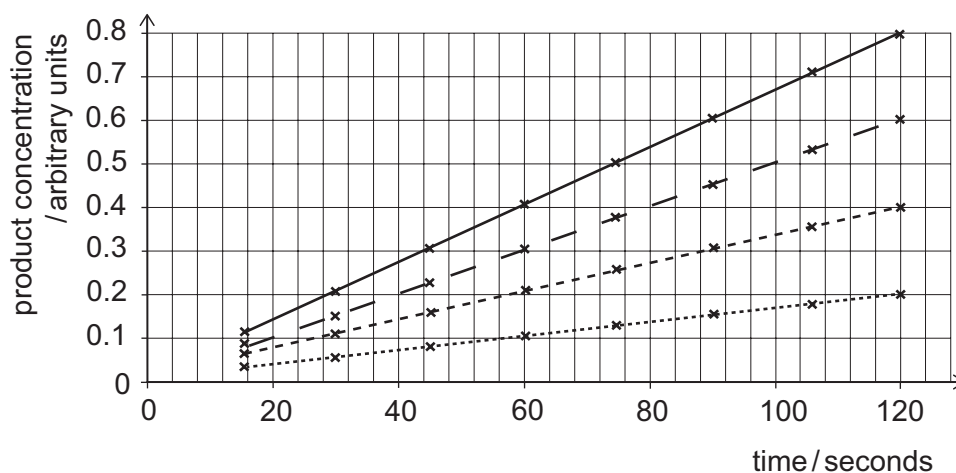
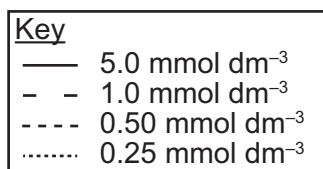


- 71 An investigation was carried out on the effect of substrate concentration on an enzyme-controlled reaction.

Four different concentrations of substrate were tested. In each case, the concentration of product was measured at regular intervals following the introduction of the substrate.

All the other variables were kept constant.

The results obtained are shown on the graph.



Which of the following statements is/are correct?

- 1 As substrate concentration increases, the time taken to produce 0.2 arbitrary units of product increases.
 - 2 Doubling the substrate concentration always doubles the rate of the reaction.
 - 3 The average rate of reaction for a substrate concentration of 1.0 mmol dm⁻³ is 0.005 arbitrary units per second.
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3